OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

PROJECT NAME

ETR1000 VZI-6

DOCUMENT TITLE

VZI-6 FRS Saloon

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1. Introduction

The purpose of this document is to provide a function specification for the control of the HVAC Saloon system for the VZI-6, extension order of V300ZEFIRO, for costumer Trenitalia.

In addition to this FRS, Hitachi Rail STS will provide and share with the client the following documents in order to assure the right functional integration of the system into the vehicle:

- System Requirement Document (SRD).
- Interface Control Document (ICD).

The SRD will include the description on general features of the system.

The ICD will include the details of the electrical and network interfaces of the system under scope to any other affected item.

1.1 Train configuration

Each VZI-6, extension order of V300ZEFIRO, trains consists of eight cars in a fixed formation including a driver's cab at each end

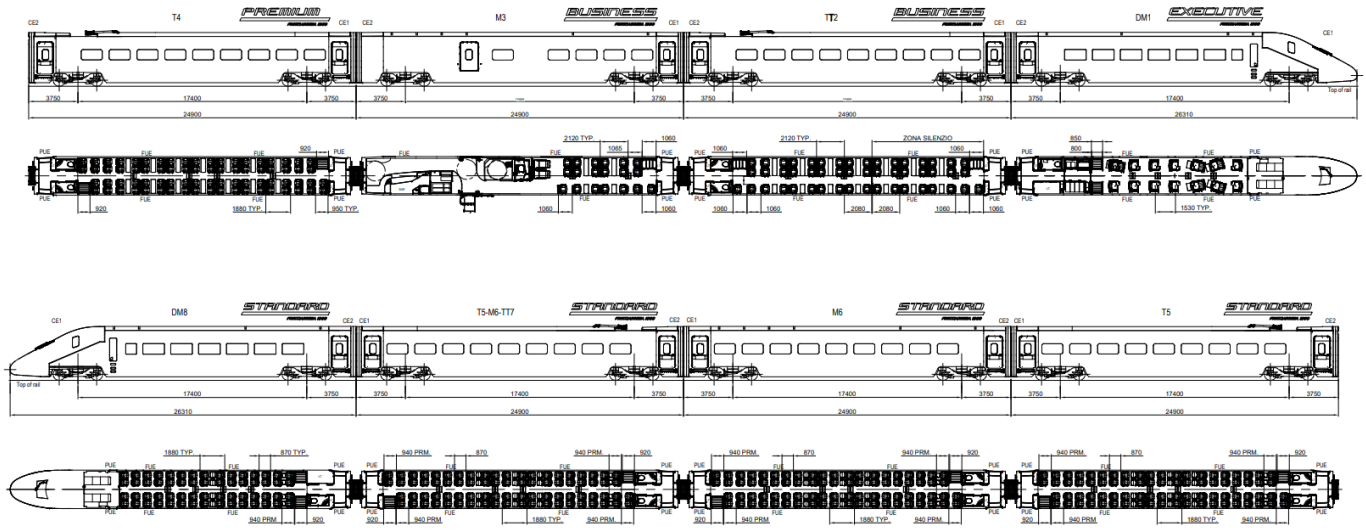


Figure 1. Train Layout

1.2 Definitions

Definition	Meaning
DM	End driving motor car with driver's cab
TT	Intermediate trailer car with transformer
M	Intermediate driving motor car
T	Intermediate trailer car

Car	Each unit carrying the passengers
Vehicle	The set of the cars which is composed of n cars and can be driven and operated independently from its both end.
System	Hardware and Software aggregate of components organized in order to realize specific functions or functions set. The term “System” identifies the entire System including the equipment and all other required and/or recommended components for system’s functions implementation.
Equipment	Set, identified by its Part Number and Serial Number, with performances specified in the system context
Component	Separate part, arrangement of parts, subsystems or units that implement a well distinct function. The term component identifies an object, with Part Number or drawing number, used in the system, but not necessarily designed for it.
Stand-Alone	Equipment and/or system working without being integrated on the vehicle.
Test Plan	• Document containing the list of the tests to be done, in sequential order and with logic connections. • The document includes the following information, related to each test: • Reference code of each test • Reference document • Reference to the requirement quoted into the Requirements’ Management • Reference to the Procedure and/or applicable Test Method • Information About series a type tests • Test execution site
Design Review	Formal, documented, exhaustive, systematic and critical review of design’s results carried out by HR – Supplier and, if requested, with the Final Customer. It allows to evaluate the design’s, and/or parts of it, fitness to the fulfilment of the input requirements considered in the design phase, identifying potential problems and the necessary corrective actions for the fulfilment of the Costumer’s requirements.
Design Freezing	Activity following from the positive result of the design review related to the considered phase or quote of the design. Design freezing consists in the definition of the design’s configuration of the reviewed design’s (or quotes of it) documentation, formalized through a document list with valid revision date (or index) following from review.

Tab. 1 - Tab. 1 - Definitions

1.3 Acronyms and abbreviations

Acronym	Meaning
BTU	Bench Test Unit
CDRL	Contractual Deliverable Requirement List
D f E	Design for Environment
DFP	Design Freeze Phase
EC	European Community

EMI	Electromagnetic Interference
FRS	Functional Requirements Specification
HR	Hitachi Rail STS
HV	High Voltage
ICD	Interface Control Document
LCA	Life Cycle Assessment
LCC	Life Cycle Cost
LRU	Last Replaceable Unit
LV	Low Voltage
MV	Medium Voltage
PDR	Preliminary Design Review
PTU	Portable Test Unit
RAMS	Reliability, Availability, Maintainability and Safety
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
TBD	To Be Defined
TCMS	Train Control and Monitoring System
TRS	Technical Requirements Specification
UNIFE	Association of the European Rail Industry
CD	Circuit Diagram
CU	Control Unit
CCU	Central Control Unit
EVENT	Diagnostics that will be stored in the Event database whiting TCMS SW
FDS	Fire Detection System
FTP	File Transfer Protocol
HVAC	Heating Ventilation and Air Conditioning
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps
LCP	Local Control Panel
LCU	Local Control Unit
IDU	Interface Display Unit
PC	Portable Computer
TCMS	Train Control and Management System

Tab. 2 - Tab. 2 – Acronyms and abbreviations

2. Reference standards and documents

2.1 Hitachi documents

Reference	Document Number	Rev	Title
[2.1]	EC01P008880B	08	Saloon HVAC-VF.CONCEPT SFRS
[2.2]	EC130019205	00	General requirement for ICD Specification
[2.3]	EC010019346	05	System Requirement Document
[2.4]	EC130021304	00	TCMS- HVAC system Interface Control Document (ICD)
[2.5]	EC130021305	00	ETR100 – TRDP signal List – ICD-HVAC Annex
[2.6]	EC13P021830B	07	ETH TRDP Specification

Tab. 3 - Tab. 4 - List of Hitachi documents

2.2 Applicable standards

The following sections define the applicable standards. They shall be assumed at the latest issue available referred to the NTP date except to as differently stated by this documentation or the documents at §2.1.

2.2.1 General standards

Reference	Document Number	Title
[3.3]	EN 50155	Railway application – Rolling stock – Electronic equipment
[3.4]	EN 50121-3-2 EN 50121-1 EN 50121-3-1	Railway applications - Electromagnetic compatibility
[3.5]	EN 60529	Degrees of protection provided by enclosures (IP code)
[3.6]	EN 50126	Railway applications - the specification and demonstration of reliability, availability, maintainability and safety (RAMS)
[3.7]	EN 50128	Railway applications - Communication, signalling and processing systems - Software for railway control and protection systems
[3.8]	EN 50657	Railways Applications. Rolling stock applications. Software on Board Rolling Stock
[3.9]	EN 61373	Railway applications. Rolling stock equipment - Shock and vibration tests
[3.10]	EN 50343	Railway applications - Rolling stock - Rules for installation of cabling
[3.11]	EN 45014	General criteria for suppliers -- Declaration of conformity
[3.12]	EN 45545 series	Railway applications - Fire protection of railway vehicles
[3.13]	EN 50153	Railway applications -- Rolling stock. Protective provisions relating to electrical hazards

[3.14]	ISO 9223	Corrosion of metals and alloys -- Corrosivity of atmospheres -- Classification
[3.15]	ISO 3381	Railway applications — Acoustics — Measurement of noise inside rail bound vehicles
[3.16]	EN 60077 series	Railway applications - electric equipment for rolling stock
[3.17]	IEC 61375	Electronic railway equipment - Train communication network (TCN)
[3.18]	EN 50215	Railway applications - rolling stock - testing of rolling stock on completion of construction and before entry into service
[3.19]	EN 50124-1	Railway applications - insulation coordination - part 1: basic requirements - clearances and creepage distances for all electrical and electronic equipment
[3.20]	EN 50500	Measurement procedures of magnetic field levels generated by electronic and electrical apparatus in the railway environment with respect to human exposure
[3.21]	EN 12663	Railway applications – Structural requirements of railway vehicle bodies – Part 1: Locomotives and passenger rolling stock (and alternative method for freight wagons)
[3.22]	EN 15227	Railway applications – Crashworthiness requirements for railway vehicle bodies
[3.23]	EN 50264	Railway applications – Railway rolling stock power and control cables having special fire performance
[3.24]	EN 50306	Railway applications – Railway rolling stock cables having special fire performance – Thin wall
[3.25]	EN 50200	Method of test for resistance to fire of unprotected small cables for use in emergency circuits
[3.26]	EN 50533	Railway applications – Three-phase train line voltage characteristics
[3.27]	ISO 5457	Technical product documentation – Sizes and layout of drawing sheets
[3.28]	ISO 10209	Technical product documentation - Vocabulary
[3.29]	ISO 5456 series	Technical drawings

Tab. 4 - Tab. 5 – List of general standards

2.2.2 System specific applicable standards

Reference	Document Number	Title
[4.1]	TSI RST (EU) 2019/776	TSI Rolling Stock
[4.2]	EN 12100-1	Safety of machinery - Basic concepts, general principles for design - Part 1: Basic terminology, methodology
[4.3]	EN 12263	Refrigerating Systems and heat pumps - Safety switching devices for limiting the pressure - Requirements and tests
[4.4]	EN 12735	Copper and copper alloys - Seamless, round copper tubes for air conditioning and refrigeration
[4.5]	EN 13129-1	Railway applications - Air conditioning for main line rolling stock – Part 1 Comfort parameters
[4.6]	EN 13129-2	Railway applications - Air conditioning for main line rolling stock – Part 2: Type tests

[4.7]	EN 14813-1	Railway applications - Air conditioning for driving cabs – Part 1 Comfort parameters
[4.8]	EN 14813-2	Railway applications - Air conditioning for driving cabs – Part 2 Type tests
[4.9]	UIC651	Layout of driver's cabs in locomotives, railcars, multiple unit trains and driving trailers
[4.10]	UIC660	Measures to ensure the technical compatibility of high speed trains
[4.11]	EN 378 P1 - P4	Refrigerating Systems and heat pumps - Safety and environmental requirements Part 1 to Part 4
[4.12]	EN 50125	Railway applications - Environmental conditions for equipment - Part 1: Equipment on board rolling stock
[4.13]	EN 50121-3-2	Railway applications – Electromagnetic compatibility - Part 3-2: Rolling stock – Apparatus
[4.14]	EN 50125-1	Railways application - Environmental conditions for equipment - Part 1: Equipment on board rolling stock
[4.15]	EN 60204-1	Safety of machinery - Electrical Equipment of machines - General requirements
[4.16]	EN 60349-2	Railway applications - Rotating electrical machines for rail and road vehicles - Part 2: Electronic converter-fed alternating current motors
[4.17]	EN 779	Particulate air filters for general ventilation - Determination of the filtration performance
[4.18]	EN ISO 3095	Railway applications – Acoustics – Measurement of noise emitted by railbound vehicles
[4.19]	EN 45545	Applicazioni ferroviarie - Protezione al fuoco per i rotabili ferroviari
[4.20]	EN 50343	Railways application - Rolling stock - Rules for installation of cabling

Tab. 5 - Tab. 6 - List of System specific applicable standards

3. System Overview

The HVAC Control architecture of one car is composed by the following main elements:

- 2 Saloon HVAC modules per coach (1 control unit inside each HVAC saloon module)
- 1 Cab HVAC control unit inside each HVAC Cab module. Only in DM1 and DM8 cars.

The redundancy of the 2 HVAC roof mounted is one of the strengths of the system. In case of failure of one controller the other of the car can take over the tasks that the first was performing. Apart from communication. There are several discrete signals that are foreseen to receive rapidly important information:

- Tunnel mode pressure transducers
- ON/OFF signal
- Internal fire signal (from detection system provided by train)

The HVAC system is arranged in such a way that each car is equipped with two HVAC controllers, the controllers are interconnected with the HVAC units via one or two CAN busses depending on car type. Solid lines represent data flows exchanged under normal conditions. The dotted lines represent the data flows exchanged in the event that the control unit that manages the data under normal conditions is faulty

For deep analisys of architecture and logic of the system control communication see [2.4], [2.5] and [2.6].

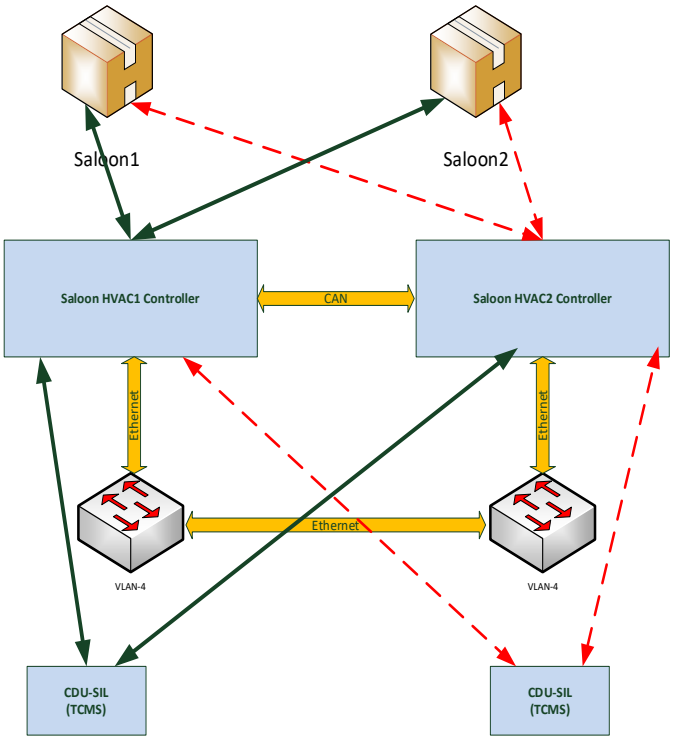


Figure 2. Saloon Controller Logic

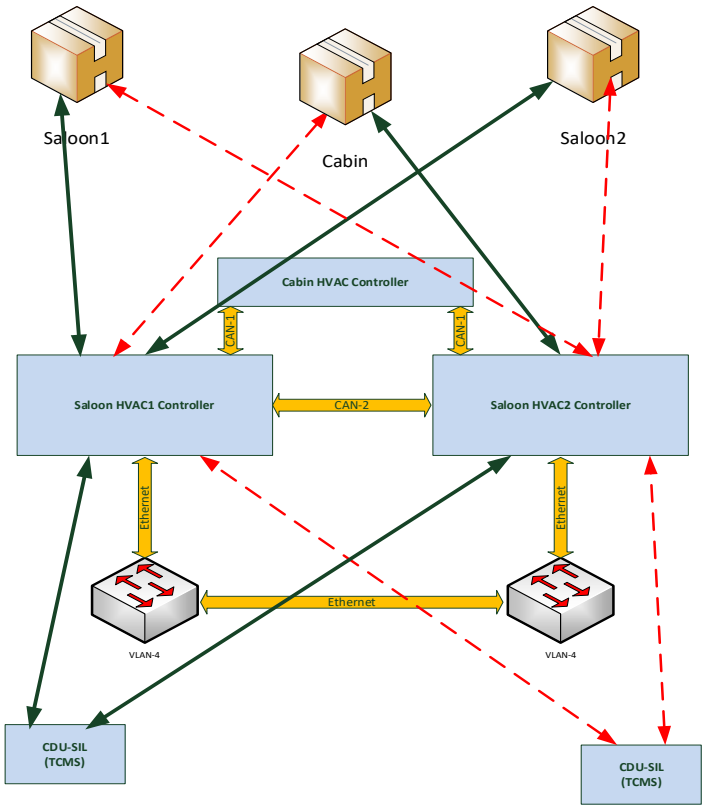


Figure 3. DM Controller Logic

4. Functional Requirements

4.1 Priority of operational mode

4.1.1 Function Design

4.1.2 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.11
Description	The safety relevant mode “fire Inside” shall have highest priority
Safety level	SIL 1

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.197	The Safety relevant mode "Fire Inside" shall have highest priority	Derived Requirement	4S_13.-.HVAC		SIL 2

ID	2F_15.02.Zefiro-Europe.TRS.99
Description	The priority order below the Fire Inside mode shall be: HVAC OFF mode Battery mode
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.117	The priority order below the Fire Inside mode shall be: HVAC OFF mode Battery mode	Derived Requirement	4S_13.-.HVAC		SIL 2

4.2 HVAC Saloon – Fire Inside mode

4.2.1 Function Design

In case of FIRE detection (*FIRE PROTECTION SYSTEM*) the HVAC system will be switched OFF by hardwired train line.

4.2.2 Normal Condition

Whenever the Fire Protection System is alarmed by FIRE event, the HVAC System of the coach where the fire is detected will be switched off by hardwired train line. The HVAC of all other coaches where fire is not present will work in full recirculation mode.

4.2.3 Failure Condition

The HVAC cannot enter fire mode when ordered.

4.2.4 Failure Supervision

A fault shall be reported: 1°) Fire Fault HVAC signal; 2°) HVAC fault code – (i.e Damper fault). The TCMS will make a comparison between the status sent by HVAC and the Fire Protection System of the same coach and a diagnostic event shall be displayed to the driver if they differ.

4.2.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.15
Description	The saloon HVAC shall be switched off by hardwired train line when fire has been detected
Safety level	Sil 1

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.198	HVAC shall shutdown and close the interface (fresh air, exhaust air) to the exterior part of the train when fire is indicated by Fire Protection System. The information about fire shall be in failsafe logic.	Derived Requirement	4S_13.-.HVAC		SIL 2
2F_15.02.Zefiro-Europe.CONCEPT.199	The HVAC shall communicate to TCMS that fire is detected (HVAC_FIRE_MODE signal). If the HVAC cannot enter fire mode when ordered a fault shall be reported · HVAC_FIRE_FAULT signal ; · HVAC fault code – (i.e Damper fault).	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.200	The TCMS shall make a comparison between the status sent by HVAC and the Fire Protection System of the same coach and a diagnostic event shall be displayed to the driver if they differ.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.565	Each HVAC unit shall work in HVAC Full Recirculation Mode as soon as it is commanded in this status by TCMS.	Derived Requirement	4S_13.-.HVAC		SIL 0

Whenever the Fire Pre – Alarm is set locally by the fire detection system, TCMS will order HVAC full Recirculation mode in the coach where pre-alarm is set (unless no other mode with higher priority are set). If real fire is confirmed by fire detection system, the local HVAC unit (of the car where the fire is detected) will be shut down by HW signal by Fire detection system.

4.3 HVAC Saloon – Off mode

4.3.1 Function Design

It will be possible to switch OFF all HVAC units in the train by a control on the driver's desk, this is typically used when there is a fire outside the train. It will also be possible for the driver to switch OFF (cut out) individual HVAC units using the IDU.

4.3.2 Normal Condition

The driver will be able to turn OFF the HVAC Saloon system per train using a control on driver's desk or individually using the IDU. Train STAFF will be able to turn OFF the HVAC unit per CAR by a Local Switch ON / OFF selector present in each car on the HVAC Local Control Panel.

4.3.3 Failure Condition

1. The HVAC system cannot enter in HVAC OFF mode by the driver using control on driver's desk
2. The HVAC system cannot enter in HVAC OFF mode by the driver using the IDU
3. The HVAC system cannot enter in HVAC OFF mode by the Train Staff

4.3.4 Failure Supervision

[Failure condition 1+2] If the HVAC Controller cannot fulfil the requested OFF mode requested by the driver using control on driver's desk it will be communicated to TCMS (General Fault HVAC, HVAC fault code). The Train Staff will have the possibility to shut down the system by Local Panel.
If the HVAC Controller cannot fulfil the requested OFF mode it will be communicated to TCMS (General Fault HVAC, HVAC fault code). The driver will have the possibility to shut down the system by IDU or control on driver's desk.

4.3.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.22
Description	The driver shall be able to shut down HVAC Saloon manually via the IDU.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.202	The driver shall be able to turn OFF and ON the HVAC Saloon individually by IDU : HVAC_OFF_MODE	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

ID	2F_15.02.Zefiro-Europe.TRS.100
Description	The driver shall be able to switch off the saloon HVAC.
Safety level	SIL 1

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.203	The driver shall have the possibility to turn off all HVAC units in the train via the drivers desk.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 1

2F_15.02.Zefiro-Europe.CONCEPT.563	The driver shall have the possibility to turn off all HVAC units in the train via the drivers desk.	Derived Requirement	4S_08.07.01.-.Drivers Desk		SIL 1

ID	2F_15.02.Zefiro-Europe.TRS.19
Description	Normal switch off (for whole train): The TCMS shall switch off the HVAC system in case of the following states (input via IDU or connected with train status): • Wash mode -> from train operational mode • Shut down -> from train operational mode (opening of battery contactor) • HVAC off order per train • HVAC off order from TCMS, if 400 Volt are not available in preconditioning mode/ frost protection mode. Local switch off (for car): • Locally switch located on the HVAC control panel (per car) • The switch could be used also to switch off the HVAC unit by crew in case of failure on communication link.
Safety level	SIL 0

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.204	When Wash mode is ordered from train operational mode, TCMS shall order the HVAC units to enter Wash mode, HVAC_WASH_MODE	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.207	If 400 Volt are not available in preconditioning / frost protection mode the HVAC system for that car shall be switched OFF.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.208	Train STAFF shall be able to turn OFF the HVAC unit per CAR by a Local Switch ON / OFF selector present in each car on the HVAC Local Control Panel even if the driver requests to switch ON the HVAC	Derived Requirement	4S_13.-.HVAC	HW HMI	N/A
2F_15.02.Zefiro-Europe.CONCEPT.209	The HVAC OFF order by the driver shall have higher priority than an ON order ordered via the HVAC Local Control Panel by train crew.	Derived Requirement	4S_13.-.HVAC		N/A

2F_15.02.Zefiro-Europe.CONCEPT.210	It shall be possible to manually turn OFF the HVAC controller by a switch positioned on HVAC Local Control Panel even if IP-network is down, (i.e TCMS cannot order HVAC OFF).	Derived Requirement	4S_13.-.HVAC	HW HMI	N/A

In order to give the HVAC unit time to be shut off in a controlled way TCMS will in the following 2 ways send a pre-warning signal to the HVAC units when the train is about to be shut off for a long time There will be a pushbutton on the IDU that can be used by the driver/staff to send a pre-warning to the HVAC units. Once the ok is given the driver will lower the pantograph and shut off the train. A pre-warning will be sent when an automatic exit from Parking mode is requested, the pre-warning will be sent a specified time in advance (10min) before the pantograph is lowered. If not all HVAC units have confirmed that it is OK to shut off an event will be set and the automatic exit from Parking mode will continue.

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.212	The driver shall have the possibility to inform the HVAC units that the train is about to be turned off via a push button on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_15.02.Zefiro-Europe.CONCEPT.447	When the HVAC unit is OK to be shut off after the pre-waring have been sent by the driver it shall be indicated to TCMS	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.448	When all HVAC units have confirmed that it is OK to shut off after the pre-waring have been sent by the driver it shall be indicated to the driver	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_15.02.Zefiro-Europe.CONCEPT.551	TCMS shall store an event if the HVAC units not have confirmed that it is OK to be shut down within an adjustable time after the pre-warning command (by driver or during parking mode exit) have been sent. The default value for the adjustable time shall be 180s.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_15.02.Zefiro-Europe.CONCEPT.569	If parking mode is pending to be automatically switched OFF a pre-warning signal shall be sent to the HVAC units	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

4.4 HVAC Saloon – Automatic mode

4.4.1 Function Design

Within the HVAC Functional Mode, the Automatic Mode is foreseen. In order to individuate graphically the process followed automatically by HVAC Controller in HVAC Automatic Mode, considering always the priority assumed for operational mode.

4.4.2 Normal Condition

The system will regulate the internal temperature in agreement to comfort requested by standard.

4.4.3 Failure Condition

1. The HVAC cannot regulate at full power the internal comfort with the next conditions. The internal comfort will be more or less degraded depending by external ambient conditions.
   a) Failure of the Ventilation system of one of the 2 HVAC saloon units.
   b) Failure of the heating system of one of the 2 HVAC saloon units when there is a situation of heating demand.
   c) Failure of cooling system of one of the 2 HVAC saloon units when there is a situation of cooling demand.
2. Complete Failure of all components of both HVAC module.

4.4.4 Failure Supervision

1. If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). Then the HVAC saloon unit in degraded mode will operate automatically in order to obtain the best comfort possible with the degraded resources available.
2. If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). The internal comfort will not be provided and the coach where there is a Global Fault (No ventilation provided; No air conditioning provided) has to be evacuated.

4.4.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.24
Description	In automatic mode shall the controller automatically switch between the modes Cooling, Ventilation or Heating. The interior temperature shall be set depending on the ambient temperature.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.213	Whenever HVAC is working in automatic conditions the HVAC shall regulate temperature in agreement to SET point curve obtained by Standards and implemented in HVAC SW.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.214	The HVAC controller shall communicate the mean internal temperature for each coach.	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.02.Zefiro-Europe.TRS.25
Description	When no other mode is requested the saloon HVAC system shall automatically operate in Automatic mode.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.215	The HVAC shall work in HVAC_AUTOMATIC_MODE when no other FUNCTIONAL mode is requested and when no other operational mode with higher priority is requested.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.216	The HVAC Controller shall communicate the actual operational mode to TCMS.	Derived Requirement	4S_13.-.HVAC		N/A

4.5 HVAC Saloon – Platform mode

4.5.1 Function Design

Whenever the train is standing in platform the TCMS shall give the signal “platform mode” to the HVAC Control Unit as represented in the figure below (Figure 4: HVAC Platform Mode- signal management).

4.5.2 Normal Condition

The system will operate in platform mode when receives the platform mode signal by TCMS reducing the air condenser fan velocity in order to reduce noise when the train is standstill in platform.

4.5.3 Failure Condition

The HVAC cannot operate in Platform mode when requested by TCMS. The noise produced by condenser Fan cannot be reduced.

4.5.4 Failure Supervision

If the HVAC Controller cannot fulfil the requested Platform mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code).

4.5.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.28
Description	As soon as any doors are released, and if no other mode with higher priority is set, all saloon HVAC systems shall operate in platform mode. As soon as there is a door close command from the driver, HVAC shall leave platform mode.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.220	Once the doors are released TCMS shall send the “HVAC_PLATFORM_MODE” signal to the HVAC Control Unit unless no other mode with higher priority are set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.222	Once the HVAC Controller receives the Platform mode signal by TCMS , the HVAC shall operate in order to reduce the noise as more as possible and in order to fulfil requirements detailed in TRD	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.223	Once all the doors on the train are closed and the HVAC Control Unit is in “HVAC_PLATFORM_MODE” TCMS shall order the HVAC in HVAC_AUTOMATIC_MODE unless no other mode with higher priority are set.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

4.6 HVAC Saloon – Tunnel mode

4.6.1 Function Design

In the picture below is shown the Tunnel mode strategy considering:

- Priority check see:
  - HVAC in case of Fire
  - HVAC OFF mode + HVAC Battery mode
- The pressure stroke disease is more important than CO2 high level
- The driver will have the possibility to set and reset Tunnel mode (activate and de-activate tunnel mode manually) via the push button (PB) located on the driver's desk
- System solves automatically the situation when driver leaves pushed the push button and doesn't remember to push it again to deactivate train line, in fact if the CO2 level will exceed the maximum limit allowed for more than an adjustable time (eg. 10 minutes) the tunnel mode train line will be deactivated and Flaps automatically will open.

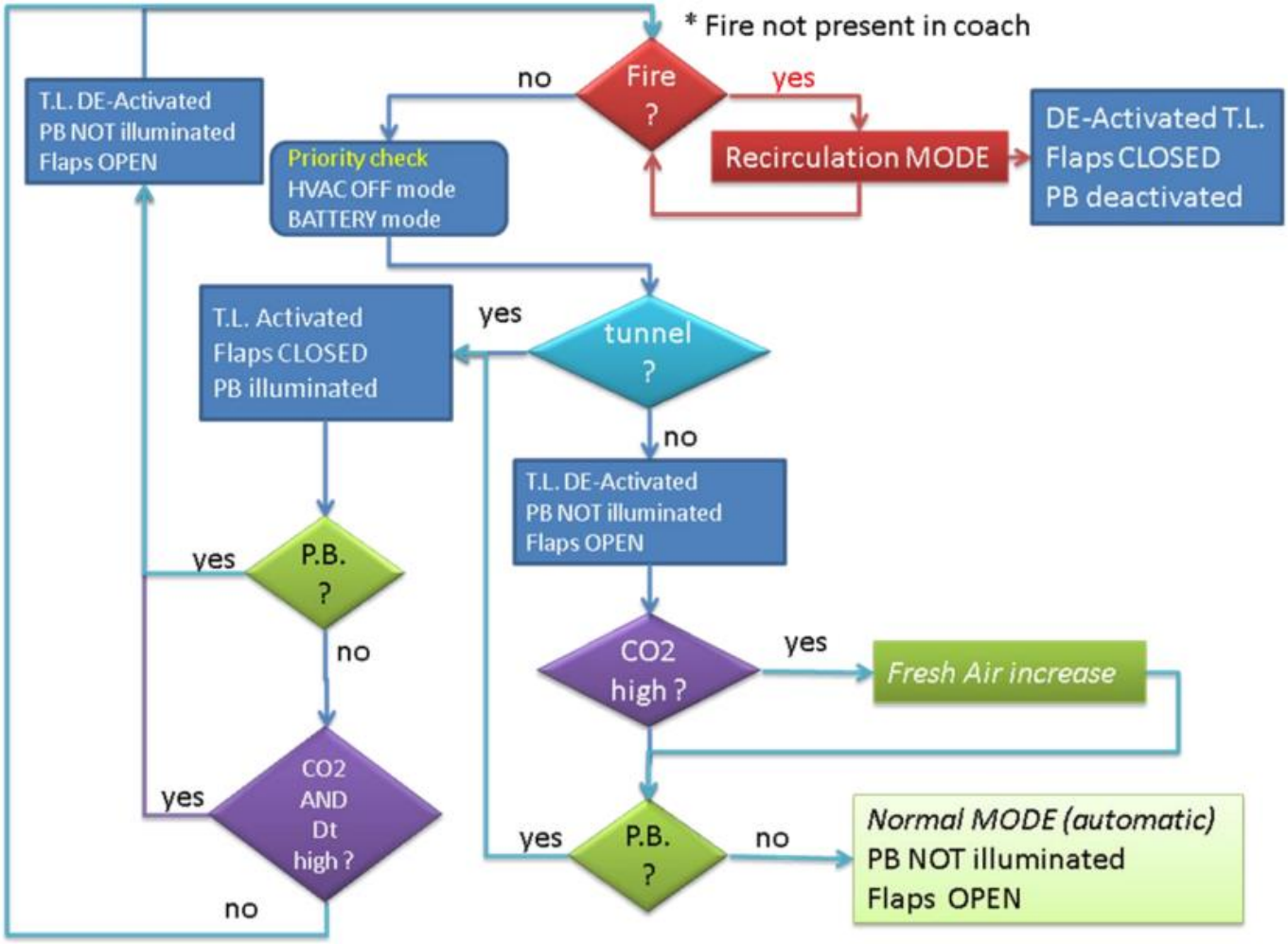


Figure 4. Signal Management

The CO2 sensors will measure the CO2 level constantly in all the coaches. When the HVAC is in Tunnel Mode and no fire is detected the HVAC prevents the CO2 level from increasing by opening the dampers after an adjustable time (eg. 10 minutes) during which the CO2 level has exceeded the maximum limit allowed. The opening of dumpers, in this scenario, is done automatically by HVAC. The pressure detection sensors will measure the pressure difference between the internal and external part of the train. The value measured out by the sensors will permit the HVAC LCU to foresee the pressure waves that can appear when entering a tunnel. It will be also possible for the driver to request (set and reset) Tunnel mode using a push button on the driver's desk. When Tunnel mode is set, either by the pressure transducers or by the driver, this is indicated by illuminating the push button on the driver's desk.

4.6.2 Normal Condition

The system will operate in Tunnel mode when requested by Driver or by pressure transducers in order to avoid that the pressure wave causes any discomfort for the passengers when entering a tunnel.

4.6.3 Failure Condition

1. The HVAC cannot operate in Tunnel mode when requested by pressure transducers.
2. The HVAC cannot operate in Tunnel mode when requested by Driver via tunnel mode push button.
3. The dampers cannot open again after HVAC Tunnel mode.

4.6.4 Failure Supervision

1. The driver will activate and deactivate Tunnel mode manually.
2. The Tunnel mode will be activated and de-activated by pressure transducers automatically.
3. The HVAC prevents the CO2 level from increasing by opening the dampers after an adjustable time (eg. 10 minutes) during which the CO2 level has exceeded the maximum limit allowed.

[Failure Condition 1+2] If the HVAC Controller cannot fulfil the requested Tunnel mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code).
[Failure condition 3] If dampers don't open after HVAC Tunnel mode, HVAC will supervise the CO2 level and it will be indicated to TCMS when the CO2 level is too high. TCMS will set a diagnostic event if CO2 level is indicated to be too high by HVAC. If damper cannot open in any conditions it will be communicated to TCMS (HVAC General Fault, HVAC fault code).

4.6.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.33
Description	When signal from differential pressure transducer is activated and if no other mode with higher priority is set, all HVAC systems shall operate in tunnel mode. The driver shall also be able to activate and de-activate tunnel mode manually.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.225	When "Tunnel mode" is indicated on the HVAC saloon unit input the HVAC saloon unit shall enter Tunnel mode to close HVAC Saloon pneumatic dampers.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.426	TCMS shall compare status from all HVAC units when "HVAC tunnel mode" is set and a diagnostic event shall be displayed to the driver if they differ.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_15.02.Zefiro-Europe.TRS.34
Description	The air inlet and outlet flaps shall be possible to close when running in tunnel mode.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.226	Whenever the HVAC is in HVAC_TUNNEL_MODE the HVAC unit shall close all the air inlet and outlet sections and all the flaps in order to avoide the pressure wave discomfort for passengers	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.227	HVAC shall supervise the CO2 level and it shall be indicated to TCMS when the CO2 level is too high.	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.442	TCMS shall set a diagnostic event if CO2 level is indicated to be too high by HVAC.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

2F_15.02.Zefiro-Europe.CONCEPT.584	TCMS shall communiucate to all HVAC Control Units (including Cab unit) that CO2 level is high in one or more coach.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL0

Once the High CO2 Level is detected higher than maximum limit for more than an adjustable time (eg 10 minutes) The HVAC Cab controller will deactivate the train line Tunnel mode even if pressure wave signal is still present.

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.458	Once the Tunnel mode Train Line is deactivated, and once the pressure discomfort is disappeared all the HVAC dampers of all the saloon units shall open	Derived Requirement	4S_13.-.HVAC		SIL 0

The driver shall have the possibility to Set or Reset Tunnel mode by a push button on the driver desk.

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.587	Once the Tunnel mode Train Line is deactivated, the HVAC Controllers of each coach shall open the fresh air dumpers.	Derived Requirement	4S_13.-.HVAC		SIL 0

4.7 HVAC Saloon – Battery mode

4.7.1 Function Design

Whenever the no *400V power supply* signal is sent by TCMS to the HVAC-LCU and when are not other higher priority mode, the HVAC operational mode will be: Battery Mode. It is possible to set Battery mode to HVAC controller by signal from TCMS.

4.7.2 Normal Condition

The system will operate in Battery mode if no *400 V power supply* signal is sent by TCMS to the HVAC-LCU.

4.7.3 Failure Condition

1. The HVAC cannot operate in Battery mode. The fresh air ventilation from external outlets cannot be provided.
2. The Battery mode is not requested to all HVAC units.

4.7.4 Failure Supervision

1. If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). If no ventilation is provided by Low Voltage supply, the CO2 level will increase because the Fresh air cannot be provided by external means. HVAC will supervise the CO2 level and it will be indicated to TCMS when the CO2 level is too high. TCMS will set a diagnostic event if CO2 level is indicated to be too high.
2. TCMS shall compare status from all HVAC units when Battery Mode is requested and a diagnostic event will be displayed to the driver if they differ.

4.7.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.38
Description	Battery mode shall be requested when - no 3AC auxiliary power available AND - no other mode with higher priority is set
Safety level	SIL 0

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.228	TCMS shall order the HVAC to HVAC_BATTERY_MODE when the following conditions are fullfilled: - no 3AC auxiliary power available AND - no other mode with higher priority is set AND none of the following modes are set - Frost protection mode - Preconditioning mode - Maintenance mode	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.443	The HVAC Controller shall go into HVAC_Battery_Mode when ordered by TCMS and no other mode with higher priority is set.	Derived Requirement	4S_13.-.HVAC		SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.229	TCMS shall compare status from all HVAC units when “HVAC_Battery_Mode” is requested and a diagnostic event shall be displayed to the driver if they differ.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

ID	2F_15.02.Zefiro-Europe.TRS.39
Description	Battery mode shall work for 30 min when battery capacity is sufficient and no other mode will be requested
Safety level	SIL 0

Whenever the HVAC system is in HVAC_Battery Mode, the evaporator fans will be supplied by batteries low voltage 110V using an inverter internal to the HVAC module.

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.459	The ventilation shall be provided by batteries for not less than 30 minutes.	Derived Requirement	4S_04.04.01.-.Batteries		SIL 0

4.8 HVAC Saloon – Maintenance mode

4.8.1 Function Design

The Ethernet Network for SW download and to perform all the service interface activities will be used. The PC will be connected to suitable switch on the network to communicate with the HVAC unit in agreement with the Network specific requirements. This mode will be activated by request of external device (e.g. PC) when the train is in the depot.

4.8.2 Normal Condition

The system will operate in Maintenance mode when a PC is linked to Ethernet connection and when maintenance activities are necessary during train depot mode.

4.8.3 Failure Condition

1. The HVAC cannot operate in Maintenance mode and no connection between PC and HVAC controller can be
2. The SW cannot be uploaded in HVAC Controller.

4.8.4 Failure Supervision

1. There is the possibility to link the PC directly to HVAC controller by RS 232 serial port inside HVAC module roof mounted.
2. The SW can be uploaded in HVAC Controller by RS 232 serial port present inside the module roof mounted.

4.8.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.43
Description	It shall be possible to set the HVAC in test mode.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.233	It shall be possible to set the HVAC controller in test mode in order to perform trouble shooting activities. It shall be possible to set the HVAC in test mode by PC through the IP network interface to TCMS for remote connection for central on-board use as well as for direct local connection when disconnected from the TCMS network.	Derived Requirement	4S_13.-.HVAC		SIL 0

4.9 HVAC Saloon – Frost protection mode

4.9.1 Function Design

It will be possible to set the HVAC unit in Frost protection mode. This will typically be used when the train is in parking mode in order to save energy. The HVAC will in Frost protection mode keep the interior temperature greater than an internal minimum level of temperature “min_int.Temp” (e.g. 5°C). At the moment Frost protection mode is not used in Zefiro Italy

4.9.2 Normal Condition

The system will operate in Frost protection mode when Parking mode signal is sent by TCMS.

4.9.3 Failure Condition

The HVAC cannot operate in Frost protection mode avoiding the protection of on board systems against low temperature conditions.

4.9.4 Failure Supervision

If the HVAC Controller cannot fulfil the requested Frost Protection mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code).

4.9.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.190
Description	It shall be possible for TCMS to set HVAC saloon in Frost protection mode
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.456	When ordered by TCMS, the HVAC unit shall enter frost protection mode.	Derived Requirement	4S_13.-.HVAC		N/A

4.10 HVAC Saloon – Communication missing mode

4.10.1 Function Design

In case of communication failure of both saloon HVAC units, the HVAC will work in normal ventilation mode until a mode with higher priority is activated by a hardware input.

4.10.2 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.52
Description	In case of communication failure of the IP link of both saloon HVAC units, the HVAC shall work in normal ventilation mode until a mode with higher priority is activated by a hardware input.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.234	The HVAC unit shall work in normal ventilation mode if TCMS-communication is lost for both saloon HVAC units AND no other mode with higher priority is set	Derived Requirement	4S_13.-.HVAC		N/A

In HVAC_COMMUNICATION_MISSING mode the HVAC provides only ventilation.

4.11 HVAC Saloon – Preconditioning mode

4.11.1 Function Design

This mode can be started only after start-up of the HVAC Saloon unit, in case the average internal temperature inside the coach is above set-point temperature. In order to reach the set-point of internal temperature as soon as possible, Fresh air inlets of the HVAC unit will be closed and thus the system will work only with return air.

Conditions to run Preconditioning are:

- ON signal activated.
- Average Internal temperature above set point.
- Ventilation running confirmation by airflow detector.
- “AC voltage available” received from TCMS.
- Cooling system working OK.

This mode ends because of one of the next reasons:

- Tim - 1ºC has been reached in passenger’s area in pre-cooling or Tim + 1ºC in pre-heating
- If there is any indication, that there are passengers on-board (e.g. required platform mode, train moving etc.)
- Maximum pre-conditioning time has elapsed
- Signal “Immediate pre-conditioning Stop” requested by TCMS.

4.11.2 Normal Condition

The system will operate in Preconditioning mode in order to reach internal target for passenger comfort.

4.11.3 Failure Condition

1. The HVAC cannot operate in Preconditioning mode and the temperature inside the coach cannot reach the internal comfort to start the train passenger service.
2. The HVAC Automatic mode will not start after preconditioning phase

4.11.4 Failure Supervision

1. If the HVAC Controller cannot fulfil the requested Preconditioning mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). The coach where the internal comfort cannot be provided can be put out of order.
2. If the HVAC Controller cannot fulfil the requested automatic mode after preconditioning phase, it will be communicated to TCMS (HVAC General Fault, HVAC fault code).

4.11.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.61
Description	The HVAC saloon system shall switch from Preconditioning mode to normal operation when - requested by the driver via the IDU OR - if there is any indication, that there are passengers on board (required platform mode, train moving etc.) OR - Preconditioning mode shall be automatically deactivated if the internal targets in terms of comfort are achieved.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.235	TCMS shall send to HVAC Controller the HVAC_PRECONDITIONING_MODE signal after start-up of the train and when line voltage is available.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.236	Whenever the driver requests to conclude the HVAC_PRECONDITIONING_MODE phase the TCMS shall set the HVAC Controller in HVAC_AUTOMATIC_MODE if no other functional mode are requested.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A
2F_15.02.Zefiro-Europe.CONCEPT.237	Whenever HVAC are in HVAC_PRECONDITIONING_MODE and doors are released (passengers are allowed to come inside the vehicle) or if there are any other indication that passengers are onboard the TCMS shall send to HVAC Controller the HVAC_PLATFORM_MODE signal if no other functional mode are requested.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.238	The HVAC shall shift from HVAC_PRECONDITIONING_MODE to HVAC_AUTOMATIC_MODE (if no other functional mode are requested) if the internal targets in terms of comfort are achieved.	Derived Requirement	4S_13.-.HVAC		N/A

4.12 HVAC Saloon – Reduced power supply mode

4.12.1 Function Design

The system will enter Reduced Power mode when reduced auxiliary power supply is available. The HVAC will not use all the power requested to reach the internal comfort required by TRD.

4.12.2 Normal Condition

The system will operate in Reduced power supply mode in order to reach internal target for passenger comfort even if a reduced power is available.

4.12.3 Failure Condition

The HVAC cannot operate in Reduced power supply mode and the temperature inside the coach cannot reach the internal comfort.

4.12.4 Failure Supervision

If the HVAC Controller cannot fulfil the requested Reduced power supply mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code).

4.12.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.63
Description	If reduced auxiliary power supply is available the saloon HVAC system shall work at reduced cooling and heating performances.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.239	HVAC Controller shall go into HVAC_REDUCED_POWER mode when reduced auxiliary power supply is available AND no other mode with higher priority is set.	Derived Requirement	4S_13.-.HVAC		N/A

The power available and current line voltage is used to individuate the best strategy to be adopted by HVAC system to fulfill the best internal comfort

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.240	TCMS shall communicate to HVAC Controller the power available per car and actual medium voltage.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.241	The HVAC Controller shall receive the HVAC_REDUCED_POWER signal by TCMS and shall work at reduced cooling and heating performances.	Derived Requirement	4S_13.-.HVAC		N/A

4.13 HVAC Saloon – Recirculation mode

4.13.1 Function Design

Whenever a fire is detected, the TCMS will visualize the Fire event on the IDU and by a relevant visual signal on the driver's desk. Whenever a fire is detected the HVAC of all other coaches where fire is not present will work in full recirculation mode, this is covered in the SFRS Fire Detection and Extinction. If the HVAC units are in Tunnel mode and fire is detected, the HVAC units in the cars where fire is not present will enter recirculation mode and the Tunnel mode push button on the driver's desk will not be illuminated and not possible to use.

4.13.2 Normal Condition

Recirculation mode is entered when requested.

4.13.3 Failure Condition

Recirculation mode cannot be entered.

4.13.4 Failure Supervision

If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). The driver can use the HVAC OFF selector on the driver's desk to turn all HVAC units OFF

4.13.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.187
Description	When fire has been detected in one car the HVAC units in the others cars shall be set in recirculation mode
Safety level	SIL 0

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.583	Recirculation mode in fire scenario shall have higher priority than Tunnel mode. When recirculation mode is requested and the HVAC unit is in Tunnel mode, recirculation mode shall be entered.	Derived Requirement	4S_13.-.HVAC		SIL 0

4.14 HVAC Saloon – Compressor start

4.14.1 Function Design

In order to avoid that different cooling compressors of the HVAC Equipment starts simultaneously, overloading the train Auxiliary Converters, compressor start-up of each saloon HVAC unit of the train set will be managed by a request-permission sequence under control of the TCMS

4.14.2 Normal Condition

HVAC controller shall send to TCMS a start request when it wants to start its compressors.

4.14.3 Failure Condition

1. some of the 4 compressors of the 2 HVAC Saloon units cannot start and it will not be possible to have full cooling power for passenger's saloon during summer.
2. All the compressors inside both the HVAC units cannot start and NO cooling power will be available. The internal comfort will be more or less degraded depending by external ambient conditions.

4.14.4 Failure Supervision

1. If the HVAC Controller cannot fulfil the requested mode it will be communicated to TCMS (HVAC General Fault, HVAC fault code). The HVAC Controller will not communicate to TCMS that the compressor is activated and the start request will be removed after a specified time.
2. After that all HVAC Controller will not communicate to TCMS that the compressors are activated. the start request will be removed after a specified time and the HVAC controller will communicate to TCMS (HVAC General Fault, HVAC fault code) in order to inform that no degraded comfort can be provided

4.14.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.70
Description	The HVAC compressors shall be started sequentially to ensure that the start-up load is not too high.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.249	HVAC controller shall send a start request when it want to start it's compressors : HVAC_COMPRESSOR_START signal	Derived Requirement	4S_13.-.HVAC		N/A

2F_15.02.Zefiro-Europe.CONCEPT.250	When the start-up request from a HVAC controller compressor is active TCMS shall send a start order to that HVAC controller compressor when no other HVAC controller compressor start-up request are active.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.251	The HVAC control compressor shall start after receiving the start-up permission from TCMS	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.252	The HVAC Controller shall communicate to TCMS that the compressor is activated and remove the start request	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.253	TCMS shall remove the start order after a specified time or when the start-up request is removed.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A

4.15 HVAC Saloon – Interface with HVAC cab

4.15.1 Function Design

The HVAC Saloon is interfaced with HVAC cab in order to allow exchange of data.

4.15.2 Normal Condition

Both HVAC controller Saloon will communicate with HVAC controller CAB in order to exchange relevant data with Train Computer. The HVAC Cab unit will provide internal comfort automatically.

4.15.3 Failure Condition

1. The communication of both HVAC controller Saloon with HVAC controller Cab is not possible and there will not be the possibility to exchange data with TCMS (See SFRS HVAC Cab).
2. The HVAC unit roof mounted will not provide internal comfort for driver for HVAC module fault

4.15.4 Failure Supervision

1. In case of communication failure of both HVAC Saloon controller the HVAC Cab will work in normal ventilation.
2. HVAC Saloon will provide nominal level of comfort in the cab in case of fault of components inside the HVAC cab unit. In this case the treated air will be provided directly by HVAC saloon ducting.

4.15.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.79
Description	The HVAC Saloon shall be interfaced with HVAC cab.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.254	The HVAC Controller of Saloon shall have the possibility to communicate all the relevant data and HVAC saloon status to the HVAC Controller of the HVAC Driver Cabin Unit	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.255	The HVAC Controller of saloon shall have the possibility to receive all the relevant data and HVAC Driving Cab status	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.02.Zefiro-Europe.TRS.72
Description	The nominal level of comfort in the cab shall be provided by HVAC Saloon in case of fault in the HVAC Cab
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.450	HVAC Saloon shall provide nominal level of comfort in the cab in case of fault in the HVAC cab.	Derived Requirement	4S_13.-.HVAC		N/A

4.16 HVAC Saloon – Two HVAC units in each car

4.16.1 Function Design

In order to have total redundancy of the HVAC system, the two HVAC Controllers of the two units mounted on each coach will have the possibility to communicate all the data and status parameters.

4.16.2 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.84
Description	The two saloon HVAC units in each car shall communicate with each other. In case one of the two saloon HVAC units fails in a car the other saloon HVAC unit shall regulate the comfort in the saloon.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.257	The two saloon HVAC units in each car shall communicate with each other. In case one of the two saloon HVAC units fails in a car the other saloon HVAC unit shall regulate the comfort in the saloon.	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.02.Zefiro-Europe.TRS.57
Description	When there is a fault in one of HVAC controller units the HVAC unit shall be controlled by the other HVAC unit controller. The comfort level according to Trenitalia CTP for failure mode shall then apply.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.258	When there is a fault in one of unit controller the unit shall be controlled by the other unit controller. The comfort level according to Trenitalia CTP for failure mode shall then apply.	Derived Requirement	4S_13.-.HVAC		N/A

4.17 HVAC Saloon – Comfort control Data

4.17.1 Function Design

By a local control panel, the train staff will have the possibility to set the Tim (mean internal temperature)

4.17.2 Normal Condition

Train-staff or the Driver will be able to set offset from the nominal set-point value in the passenger area respectively via the local control panel or via IDU. The external reading temperature sensors will communicate to TCMS the external temperature in order to allow the calculation of the mean exterior temperature that will be send back to HVAC Controllers in order to regulate the internal comfort considering only one

4.17.3 Failure Condition

1. The Local Control panel cannot communicate to HVAC Saloon Controller in order to change the offset Set point for the internal temperature.

2. The Driver cannot communicate to relative HVAC Controller by IDU in order to change the offset Set point for the internal temperature.
3. The external reading temperature sensors cannot communicate to TCMS the external temperature.

4.17.4 Failure Supervision

1. It will be possible to change Set Point by driver. If even the driver will not be able to change the internal Set point the internal temperature target will remain the temperature requested by standards.
2. It will be possible to change Set Point by Train Staff. If even the Train Staff will not be able to change the internal Set point the internal temperature target will remain the temperature requested by standards.
3. In the calculation of the mean exterior temperature will be taken into account only feasible values.

4.17.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.66
Description	Train-staff shall be able to set offset from the nominal set-point value in the passenger area via the local control panel. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K). The offset shall be possible to set independently for each car.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.259	Train-staff shall be able to set offset from the nominal set-point value in the passenger area via the local control panel. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K). The offset shall be possible to set independently for each car.	Derived Requirement	4S_13.-.HVAC	HW HMI	N/A

ID	2F_15.02.Zefiro-Europe.TRS.71
Description	The driver or train-staff shall be able to set the temperature offset via the IDU. The temperature offset shall be set at train level. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K).
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.260	The driver or train-staff shall be able to set the temperature offset via the IDU. The temperature offset shall be set at train level. There shall be 5 offsets (-2 K, -1 K, 0, +1 K, +2 K).	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

2F_15.02.Zefiro-Europe.CONCEPT.261	The HVAC local command panel shall have the priority with respect to any temperature requests by the driver via IDU	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.02.Zefiro-Europe.TRS.78
Description	The saloon HVAC controllers shall send the outdoor air temperatures to TCMS. The TCMS shall calculate the average outdoor air temperature based on all valid temperatures received from the HVAC controllers. The calculated average outdoor air temperature shall be sent back to the HVAC controllers.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.262	The saloon HVAC controllers shall send the outdoor air temperatures to TCMS.	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.263	The TCMS shall calculate the average outdoor air temperature based on all valid temperatures received from the HVAC controllers. The maximum value received shall not be considered when calculating the global mean value of the external temperature (this in order to exclude sensors directly affected by sun).	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.264	The calculated average outdoor air temperature shall be sent back by TCMS to the HVAC controllers.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.265	The HVAC Controllers shall receive the outside average temperature by TCMS and shall use this average temperature during regulation.	Derived Requirement	4S_13.-.HVAC		N/A

4.18 HVAC Saloon – Condition data

4.18.1 Function Design

The status of the HVAC system will be indicated on the IDU.

4.18.2 Normal Condition

The HVAC Saloon controller shall send to TCMS the Status of the system

4.18.3 Failure Condition

The HVAC Saloon controller cannot communicate the status of the system

4.18.4 Failure Supervision

In case of communication failure, a relevant error message will be shown to the driver and the system will work in “Auto” mode.

4.18.5 Requirements

ID	2F_15.02.Zefiro-Europe.TRS191
Description	The status of the HVAC system shall be indicated on the IDU.
Safety level	SIL 0

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.581	The HVAC saloon Controller shall send to TCMS the complete diagnostic of the saloon HVAC systems	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.582	The TCMS shall receive all the Diagnostic Status of saloon HVAC and shall communicate all data available by IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_15.02.Zefiro-Europe.CONCEPT.217	The operational modes "On", "Off", "Cut out" and "Faulty" shall be displayed to the driver.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	N/A

4.19 HVAC Saloon – Fault Supervision

4.19.1 Function Design

All the Fault revealed by HVAC Controller will be communicated to TCMS in order to provide the Diagnostic Status of the HVAC Systems to driver and Train Staff.

4.19.2 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.82
Description	The HVAC shall supervise itself and a fault indication shall be set in case of component faults.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.266	The HVAC Controller shall supervise itself and a fault indication shall be sent to TCMS in case of component faults.	Derived Requirement	4S_13.-.HVAC		N/A

ID	2F_15.02.Zefiro-Europe.TRS.87
Description	Complete diagnostic of the saloon HVAC system, including predictive diagnostic functions and CBM, shall be fully implemented in each saloon HVAC unit controller. The data shall be transmitted to the TCMS via the saloon HVAC bus interface.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.267	The HVAC Controller shall send to TCMS the complete diagnostic of the saloon HVAC systems	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.268	The TCMS shall receive all the Diagnostic Status of each HVAC and store it in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_15.02.Zefiro-Europe.CONCEPT.218	If the HVAC Controller cannot fulfill the requested mode it shall be communicated to TCMS (HVAC_GENERAL_FAULT, HVAC fault code).	Derived Requirement	4S_13.-.HVAC		N/A
2F_15.02.Zefiro-Europe.CONCEPT.463	TCMS shall store an event when the HVAC_GENERAL_FAULT, HVAC fault code is received from the HVAC Controller	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0

4.20 Events

4.20.1 Function Design

4.20.2 Requirements

ID	2F_15.02.Zefiro-Europe.TRS.184
Description	Diagnostics from the supplier shall be stored in the train diagnostic system.
Safety level	N/A

Nr	Description	Type	Derive to	User interface	SIL
2F_15.02.Zefiro-Europe.CONCEPT.158	Diagnostics from the supplier shall be stored in the train diagnostic system.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A

5. Signal Interface

5.1 I/O interface

5.1.1 Supplier System

Inputs:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.269	HVAC tunnel mode.	L	4S_09.03.05.-.TCMS Software
2F_15.02.Zefiro-Europe.CONCEPT.270	Fire Detected	L	2F_15.02.Zefiro-Europe.CONCEPT.198 2F_15.02.Zefiro-Europe.CONCEPT.203

Outputs:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.29			

5.1.2 TCMS SW

Inputs:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.30			

Outputs:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.31			

5.2 Bus Interface

Supplier to TCMS SW:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.271	HVAC_FIRE_FAULT	L	2F_15.02.Zefiro-Europe.CONCEPT.199
2F_15.02.Zefiro-Europe.CONCEPT.272	HVAC_GENERAL_FAULT	L	2F_15.02.Zefiro-Europe.CONCEPT.218
2F_15.02.Zefiro-Europe.CONCEPT.273	HVAC fault code	A	2F_15.02.Zefiro-Europe.CONCEPT.199 2F_15.02.Zefiro-Europe.CONCEPT.218
2F_15.02.Zefiro-Europe.CONCEPT.274	HVAC_OPERATIONAL_MODE	A	2F_15.02.Zefiro-Europe.CONCEPT.199 2F_15.02.Zefiro-Europe.CONCEPT.216 2F_15.02.Zefiro-Europe.CONCEPT.426
2F_15.02.Zefiro-Europe.CONCEPT.275	HVAC_MEAN_INT_TEMP	A	2F_15.02.Zefiro-Europe.CONCEPT.214
2F_15.02.Zefiro-Europe.CONCEPT.276	HVAC OK to shut OFF	L	2F_15.02.Zefiro-Europe.CONCEPT.447
2F_15.02.Zefiro-Europe.CONCEPT.449	CO2 level too high	L	2F_15.02.Zefiro-Europe.CONCEPT.227
2F_15.02.Zefiro-Europe.CONCEPT.277	HVAC fire detected	L	2F_15.02.Zefiro-Europe.CONCEPT.199
2F_15.02.Zefiro-Europe.CONCEPT.278	HVAC_COMPRESSOR_START request	L	2F_15.02.Zefiro-Europe.CONCEPT.249
2F_15.02.Zefiro-Europe.CONCEPT.279	HVAC_OUT_TEMP	A	2F_15.02.Zefiro-Europe.CONCEPT.262

TCMS SW to Supplier:

Nr	Description	Type	Derive to
2F_15.02.Zefiro-Europe.CONCEPT.280	Operation Mode request	A	2F_15.02.Zefiro-Europe.CONCEPT.202 2F_15.02.Zefiro-Europe.CONCEPT.228 2F_15.02.Zefiro-

			Europe.CONCEPT.235 2F_15.02.Zefiro-Europe.CONCEPT.237 2F_15.02.Zefiro-Europe.CONCEPT.456 2F_15.02.Zefiro-Europe.CONCEPT.565
2F_15.02.Zefiro-Europe.CONCEPT.282	IAuxPowerAvail	A	2F_15.02.Zefiro-Europe.CONCEPT.240
2F_15.02.Zefiro-Europe.CONCEPT.283	HVAC_COMPRESSOR_START order	L	2F_15.02.Zefiro-Europe.CONCEPT.250
2F_15.02.Zefiro-Europe.CONCEPT.284	HVAC_AVER OUT_TEMP	A	2F_15.02.Zefiro-Europe.CONCEPT.264
2F_15.02.Zefiro-Europe.CONCEPT.285	Reset “close all the HVAC dumper”	L	
2F_15.02.Zefiro-Europe.CONCEPT.286	Interior temperature set-point	A	2F_15.02.Zefiro-Europe.CONCEPT.260
2F_15.02.Zefiro-Europe.CONCEPT.287	Shutdown pre-warning	L	2F_15.02.Zefiro-Europe.CONCEPT.212

6. References

[1] EC01P006696B TRD Passenger saloon and driving cab HVAC system

[2] 3EGH000035-1410 Network specific requirements

[3] EC01P008884B SFRS Fire Detection and Extinction

[4] EC01P008880B Saloon HVAC-VF.CONCEPT SFRS